

claude-windows / b13673cc-f087-43b1-9527-35dc0b67b301

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\b13673cc-f087-43b1-9527-35dc0b67b301\subagents\workflows\wf_61401384-db8\agent-aaa444d0ef50828bd.jsonl`  
- SHA-256: `0355ea8c6d3984d203e1828dd7b5e30e1352c193f05e1bb5fc955570adc61c3d`  
- Source modified: `2026-06-06T06:48:28+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_61401384-db8`  
- session_id: `b13673cc-f087-43b1-9527-35dc0b67b301`
```

Transcript

1. user (2026-06-06T06:45:13.257Z)

You are an adversarial data-quality auditor. The user must NOT be told the data is fine if it is not, and must NOT be told it is broken if it is actually fine. Verify every hypothesis independently by parsing the files yourself (write your own python; do not trust my numbers).

KEY PATHS (all local, absolute, read with the Read/Bash/Grep tools):

```
- Roora delivery root: "C:\Users\avidu\Projects\Annotation Setup\6 June 2026"  
  - "badminton" = PASCAL VOC export (Annotations\frame_XXXXXX.xml per sampled frame, ImageSets\, labelmap.txt)  
    - "badminton (1)\annotations\instances_Test.json" = COCO export  
    - "badminton (2)\annotations.xml" = CVAT-for-video native export (has <meta><job> with frame_filter step, stop_frame, labels)  
- Roora spec PDFs the user sent the vendor:  
  - "C:\Users\avidu\Projects\Annotation Setup\Roora Annotation Brief ? Rally Golden Set (Format Spec, Scope, QA).pdf"  
  - "C:\Users\avidu\Projects\Annotation Setup\Illustrated Annotation Guide ? Rally Labeling (Supplement).pdf"  
- The three sample videos given: "G:\My Drive\Sharing\Annotation Videos\" (Video 1 - Badminton.MP4, Video 2 - Pickleball.mp4, Video 3 - Tabletennis.mp4)  
- Indexer repo: C:\Users\avidu\Projects\badminton-highlight-indexer  
- Collector repo: C:\Users\avidu\Projects\sports-data-collector  
- python available as 'python' (3.13); ffmpeg available.
```

FACTS ALREADY ESTABLISHED (verify, don't just trust; correct me if any is wrong):

```
- ffmpeg on the on-disk videos: Badminton = 5312x2988, 29.97fps, 100.7s, 3019 frames.  
Pickleball = 3840x2160, 59.94fps, ~97s. Tabletennis = 3840x2160, 59.94fps, ~58s.  
- CVAT job meta (badminton (2)/annotations.xml): stop_frame=21270, frame_filter step=30, size=710 sampled frames, annotation canvas 1920x1080, mode=interpolation, label="rally".  
- All three folders appear to be the SAME badminton job exported in 3 formats (COCO bbox [1134.1,434.73,152.19,270.73] == VOC rally-2 box xmin1134.1/ymin434.73/xmax1286.29/ymax705.46). Pickleball & table-tennis are NOT in the delivery.  
- The rally fields delivered (as CVAT object attributes, attached to a per-player bounding box): rally_number, start_time, end_time, shots_count, ending_reason, score_before (A-B), score_after (A-B), notes, rotation.  
- VOC has 24 distinct rally_numbers; 281/710 frames contain >=1 rally object; COCO has 318 annotations over 710 images.  
- HYPOTHESIS to verify: start_time/end_time attributes == source_frame/900 (e.g. rally 24 frame 19920->20160 gives 22.13->22.4; 19920/900=22.13). i.e. attribute times are off by the sampling-step factor and the TRUE temporal signal is the per-object FRAME RANGE, not the start_time/end_time attributes.  
- HYPOTHESIS: the annotated badminton source (21270 frames, 1080p) is NOT the on-disk Video 1 - Badminton.MP4 (3019 frames, 5312x2988).  
- ending_reason values used: forced_error, unforced_error, service_fault, let, other, winner.
```

TASK ? independently verify and quantify:

1. The time formula: parse the CVAT-video and/or VOC, and for every rally compare start_time/end_time attributes against the object's frame range. Is attr_time == frame/900 exactly? Compute residuals. State the true relationship.
2. The true temporal signal & its resolution given step=30 sampling.
3. Frame-count reconciliation: is there ANY fps/interpretation under which the on-disk badminton video (3019 frames, 100.7s, 29.97fps per ffprobe) could equal the CVAT 21270-frame source? Run ffprobe yourself. Conclude same-video or different-video, with arithmetic.
4. Resolution mismatch (1920x1080 anno canvas vs 5312x2988 source) and what it means for reusing the boxes.
5. Are VOC, COCO, and CVAT-video byte-for-byte the same rally data? Spot-check.
6. Rally count; overlaps between consecutive rallies' frame ranges (quantify how many overlap and by how much); gaps; is rally_number monotonic with frame order?
7. shots_count vs real duration sanity ? convert frame ranges to seconds at the plausible true fps and flag physically-impossible shot rates.
8. ending_reason histogram and compliance vs spec set {winner, into_net, out, service_fault, let, other}.

Return a precise, evidence-backed defect list with severities and a usable-as-GT verdict.

2. assistant (2026-06-06T06:45:17.327Z)

I'll start by independently verifying each hypothesis. Let me first explore the directory structure and examine the actual files.

3. user (2026-06-06T06:45:18.492Z)

```
total 8
drwxr-xr-x 1 avidu 197609 0 Jun  6 11:24 .
drwxr-xr-x 1 avidu 197609 0 Jun  6 11:20 ..
drwxr-xr-x 1 avidu 197609 0 Jun  6 11:23 badminton
drwxr-xr-x 1 avidu 197609 0 Jun  6 11:24 badminton (1)
drwxr-xr-x 1 avidu 197609 0 Jun  6 11:24 badminton (2)
```

4. user (2026-06-06T06:45:20.194Z)

```
total 232
drwxr-xr-x 1 avidu 197609      0 Jun  6 11:24 .
drwxr-xr-x 1 avidu 197609      0 Jun  6 11:24 ..
-rw-r--r-- 1 avidu 197609 232542 Jun  6 11:24 annotations.xml
---VOC---
Annotations
ImageSets
labelmap.txt
---COCO---
instances_Test.json
```

5. assistant (2026-06-06T06:45:23.101Z)

Let me examine the three annotation files in detail. I'll write Python to parse them independently.

6. user (2026-06-06T06:45:24.536Z)

```
1      <?xml version="1.0" encoding="utf-8"?>
2      <annotations>
3          <version>1.1</version>
4          <meta>
5              <job>
6                  <id>4075945</id>
7                  <size>710</size>
8                  <mode>interpolation</mode>
9                  <overlap>0</overlap>
10                 <bugtracker></bugtracker>
11                 <created>2026-06-04 06:07:18.873191+00:00</created>
```

```

12      <updated>2026-06-05 09:47:11.609271+00:00</updated>
13      <subset>Test</subset>
14      <start_frame>0</start_frame>
15      <stop_frame>21270</stop_frame>
16      <frame_filter>step=30</frame_filter>
17      <segments>
18          <segment>
19              <id>4074196</id>
20              <start>0</start>
21              <stop>709</stop>
22
23      <url>https://app.cvat.ai/api/jobs/4075945/dataset/export/api/jobs/4075945</url>
24      </segment>
25      </segments>
26      <owner>
27          <username>shahanap960</username>
28          <email>shahanap960@gmail.com</email>
29      </owner>
30      <assignee></assignee>
31      <labels>
32          <label>
33              <name>rally</name>
34              <color>#ff007c</color>
35              <type>any</type>
36              <attributes>
37                  <attribute>
38                      <name>rally_number</name>
39                      <mutable>False</mutable>
40                      <input_type>number</input_type>
41                      <default_value>1</default_value>
42                      <values>1
43                      100
44                      1</values>
45                  </attribute>
46                  <attribute>
47                      <name>start_time</name>
48                      <mutable>False</mutable>
49                      <input_type>text</input_type>
50                      <default_value></default_value>
51                      <values></values>
52                  </attribute>
53                  <attribute>
54                      <name>end_time</name>
55                      <mutable>False</mutable>
56                      <input_type>text</input_type>
57                      <default_value></default_value>
58                      <values></values>
59                  </attribute>
60                  <attribute>
61                      <name>shots_count</name>
62                      <mutable>False</mutable>
63                      <input_type>number</input_type>
64                      <default_value>0</default_value>
65                      <values>0
66                      100
67                      1</values>
68                  </attribute>
69                  <attribute>
70                      <name>ending_reason</name>
71                      <mutable>False</mutable>
72                      <input_type>select</input_type>
73                      <default_value>forced_error</default_value>
74                      <values>forced_error
75                      unforced_error
76                      service_fault

```

```

76     let
77     other
78     winner</values>
79         </attribute>
80         <attribute>
81             <name>score_before (A-B)</name>
82             <mutable>False</mutable>
83             <input_type>text</input_type>
84             <default_value></default_value>
85             <values></values>
86         </attribute>
87         <attribute>
88             <name>score_after (A-B)</name>
89             <mutable>False</mutable>
90             <input_type>text</input_type>
91             <default_value></default_value>
92             <values></values>
93         </attribute>
94         <attribute>
95             <name>notes</name>
96             <mutable>False</mutable>
97             <input_type>text</input_type>
98             <default_value></default_value>
99             <values></values>
100        </attribute>
101    </attributes>
102    </label>
103    </labels>
104    </job>
105    <dumped>2026-06-05 09:50:24.702130+00:00</dumped>
106    <original_size>
107        <width>1920</width>
108        <height>1080</height>
109    </original_size>
110    </meta>
111    <image id="0" name="frame_000000" width="1920" height="1080">
112    </image>
113    <image id="30" name="frame_000030" width="1920" height="1080">
114    </image>
115    <image id="60" name="frame_000060" width="1920" height="1080">
116    </image>
117    <image id="90" name="frame_000090" width="1920" height="1080">
118    </image>
119    <image id="120" name="frame_000120" width="1920" height="1080">
120    </image>

```

7. user (2026-06-06T06:45:25.648Z)

```

frame_000000.xml
frame_000030.xml
frame_000060.xml
frame_000090.xml
frame_000120.xml
frame_000150.xml
frame_000180.xml
frame_000210.xml
frame_000240.xml
frame_000270.xml
frame_000300.xml
frame_000330.xml
frame_000360.xml
frame_000390.xml
frame_000420.xml
frame_000450.xml
frame_000480.xml

```

```
frame_000510.xml
frame_000540.xml
frame_000570.xml
count:
710
```

8. assistant (2026-06-06T06:45:30.017Z)

Now let me look at an image element in the CVAT-video file that actually has box/rally objects with attributes. Let me find them.

9. user (2026-06-06T06:45:31.966Z)

```
122:   <box label="rally" source="manual" occluded="0" xtl="1134.10" ytl="434.73" xbr="1286.29"
ybr="705.46" z_order="0">
123:     <attribute name="rally_number">2</attribute>
124:     <attribute name="start_time">2</attribute>
125:     <attribute name="end_time">2.1</attribute>
126:     <attribute name="shots_count">2</attribute>
```

[Showing results with pagination = limit: 5]

10. assistant (2026-06-06T06:45:35.144Z)

Interesting - this box appears under `

11. user (2026-06-06T06:45:37.472Z)

```
num <image tags: 710
num <track tags: 0
num <box tags: 318
num <polygon: 0
num <points: 0
len: 232542
```

12. assistant (2026-06-06T06:45:41.643Z)

This is critical: the "CVAT-for-video native export" actually uses `image annotation export format, not the interpolation/track format. Despite `mode=interpolation` in meta, the actual data is per-image boxes. So each box lives under a specific frame image. Let me write a comprehensive parser.

13. user (2026-06-06T06:45:45.881Z)

```
total boxes: 318
distinct rally_numbers: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
21, 22, 23, 24]
count distinct: 24
frames with >=1 box: 281
```

14. assistant (2026-06-06T06:45:49.047Z)

Confirmed: 318 boxes, 24 distinct rally numbers, 281 frames with boxes. Now let me build the full per-rally analysis: frame ranges, start_time/end_time, and the time formula.

15. user (2026-06-06T06:45:55.841Z)

```
rally | nboxes | fmin | fmax | start_time | end_time | shots | ending_reason | score_before |
score_after
  1 | 37 | 180 | 1260 | 0.2 | 1.4 | 15 | other | |
ATTR_VARIES!
```

2	44	150	1950	2	2.1	2	forced_error		
3	14	2670	3060	2.9	3.4	5	unforced_error		
4	11	3720	4020	4.13	4.14	4	unforced_error		
5	9	4320	4560	4.8	5.06	4	unforced_error		
6	13	4980	5340	5.53	5.9	4	winner		
7	20	5730	6690	6.36	6.66	2	winner		
ATTR_VARIES!									
8	4	7080	7170	7.86	7.96	1	service_fault		
9	10	7590	7860	8.43	8.73	4	winner		
10	7	8280	8460	9.2	9.4	2	unforced_error		
11	10	8970	9240	9.96	10.23	3	winner		
12	5	9780	9900	10.86	10.96	1	service_fault		
13	10	10410	10680	11.56	11.86	2	winner		
14	7	10980	11160	12.2	12.4	3	winner		
15	8	11550	11760	12.83	13.06	3	forced_error		
16	26	12330	13380	13.7	13.93	2	winner		
ATTR_VARIES!									
17	4	14160	14250	15.73	15.73	1	winner		
18	19	14580	15120	16.2	16.8	9	winner		
19	6	15750	15900	17.5	17.66	2	winner		
20	21	16230	16830	18.03	18.7	10	other		
21	6	17670	17820	19.63	19.8	2	winner		
22	9	18480	18720	20.53	20.8	4	other		
23	9	19260	19500	21.4	21.66	3	winner		
24	9	19920	20160	22.13	22.4	3	winner		

16. assistant (2026-06-06T06:46:00.472Z)

Now I have the full picture. Several rallies have ATTR_VARIES - the attributes are not constant across boxes within a rally. That's important. Now let me test the time formula rigorously: is `start\_time == fmin/900` and `end\_time == fmax/900`? Let me compute residuals against multiple divisors.

17. user (2026-06-06T06:46:08.584Z)

Testing attr_time == frame/900 hypothesis

rally	fmin	st	fmin/900	resid_start	fmax	et	fmax/900	resid_end
1	180	0.20	0.200	+0.0000	1260	1.40	1.400	+0.0000
2	150	2.00	0.167	+1.8333	1950	2.10	2.167	-0.0667
3	2670	2.90	2.967	-0.0667	3060	3.40	3.400	+0.0000
4	3720	4.13	4.133	-0.0033	4020	4.14	4.467	-0.3267
5	4320	4.80	4.800	+0.0000	4560	5.06	5.067	-0.0067
6	4980	5.53	5.533	-0.0033	5340	5.90	5.933	-0.0333
7	5730	6.36	6.367	-0.0067	6690	6.66	7.433	-0.7733
8	7080	7.86	7.867	-0.0067	7170	7.96	7.967	-0.0067
9	7590	8.43	8.433	-0.0033	7860	8.73	8.733	-0.0033
10	8280	9.20	9.200	+0.0000	8460	9.40	9.400	+0.0000
11	8970	9.96	9.967	-0.0067	9240	10.23	10.267	-0.0367
12	9780	10.86	10.867	-0.0067	9900	10.96	11.000	-0.0400
13	10410	11.56	11.567	-0.0067	10680	11.86	11.867	-0.0067
14	10980	12.20	12.200	+0.0000	11160	12.40	12.400	+0.0000
15	11550	12.83	12.833	-0.0033	11760	13.06	13.067	-0.0067
16	12330	13.70	13.700	+0.0000	13380	13.93	14.867	-0.9367
17	14160	15.73	15.733	-0.0033	14250	15.73	15.833	-0.1033
18	14580	16.20	16.200	+0.0000	15120	16.80	16.800	+0.0000
19	15750	17.50	17.500	+0.0000	15900	17.66	17.667	-0.0067
20	16230	18.03	18.033	-0.0033	16830	18.70	18.700	+0.0000
21	17670	19.63	19.633	-0.0033	17820	19.80	19.800	+0.0000
22	18480	20.53	20.533	-0.0033	18720	20.80	20.800	+0.0000
23	19260	21.40	21.400	+0.0000	19500	21.66	21.667	-0.0067
24	19920	22.13	22.133	-0.0033	20160	22.40	22.400	+0.0000

start resid: mean=0.07097 max_abs=1.83333
end resid: mean=-0.09833 max_abs=0.93667

18. assistant (2026-06-06T06:46:14.124Z)

The formula `attr_time == frame/900` holds tightly (residuals are just rounding to 2 decimals, `|resid| < 0.01`) EXCEPT where the attributes vary within the rally. The big residuals (rally 2 start, rally 4/7/16 end) are exactly the ATTR_VARIES rallies ? the attribute is taken from one box but fmin/fmax come from a different box. Let me dig into those varying-attribute rallies to confirm the formula matches the per-box frame, and check whether boxes from DIFFERENT rallies are bleeding into each other (overlaps).

19. user (2026-06-06T06:46:19.769Z)

=== rally 1 ===

frame	180	frame/900=	0.200	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	210	frame/900=	0.233	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	240	frame/900=	0.267	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	270	frame/900=	0.300	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	300	frame/900=	0.333	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	330	frame/900=	0.367	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	360	frame/900=	0.400	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	390	frame/900=	0.433	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	420	frame/900=	0.467	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	450	frame/900=	0.500	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	480	frame/900=	0.533	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	510	frame/900=	0.567	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	540	frame/900=	0.600	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	570	frame/900=	0.633	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	600	frame/900=	0.667	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	630	frame/900=	0.700	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	660	frame/900=	0.733	start_time=	0.2	end_time=	1.4	shots=	1.5	reason=	other
frame	690	frame/900=	0.767	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	720	frame/900=	0.800	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	750	frame/900=	0.833	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	780	frame/900=	0.867	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	810	frame/900=	0.900	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	840	frame/900=	0.933	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	870	frame/900=	0.967	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	900	frame/900=	1.000	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	930	frame/900=	1.033	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	960	frame/900=	1.067	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	990	frame/900=	1.100	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1020	frame/900=	1.133	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1050	frame/900=	1.167	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1080	frame/900=	1.200	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1110	frame/900=	1.233	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1140	frame/900=	1.267	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1170	frame/900=	1.300	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1200	frame/900=	1.333	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1230	frame/900=	1.367	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other
frame	1260	frame/900=	1.400	start_time=	0.2	end_time=	1.4	shots=	15	reason=	other

=== rally 2 ===

frame	150	frame/900=	0.167	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	180	frame/900=	0.200	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	210	frame/900=	0.233	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	240	frame/900=	0.267	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	270	frame/900=	0.300	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	300	frame/900=	0.333	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	330	frame/900=	0.367	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error
frame	360	frame/900=	0.400	start_time=	2	end_time=	2.1	shots=	2	reason=	forced_error

reason=forced_error								
frame	390	frame/900=	0.433	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	420	frame/900=	0.467	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	450	frame/900=	0.500	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	480	frame/900=	0.533	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	510	frame/900=	0.567	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	540	frame/900=	0.600	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	570	frame/900=	0.633	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	600	frame/900=	0.667	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	630	frame/900=	0.700	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	660	frame/900=	0.733	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	690	frame/900=	0.767	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	720	frame/900=	0.800	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	750	frame/900=	0.833	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	780	frame/900=	0.867	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	810	frame/900=	0.900	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	840	frame/900=	0.933	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	870	frame/900=	0.967	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	900	frame/900=	1.000	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	930	frame/900=	1.033	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	960	frame/900=	1.067	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	990	frame/900=	1.100	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1020	frame/900=	1.133	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1050	frame/900=	1.167	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1080	frame/900=	1.200	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1110	frame/900=	1.233	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1140	frame/900=	1.267	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1170	frame/900=	1.300	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1200	frame/900=	1.333	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1230	frame/900=	1.367	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1260	frame/900=	1.400	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1800	frame/900=	2.000	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								
frame	1830	frame/900=	2.033	start_time=	2	end_time=	2.1	shots= 2
reason=forced_error								

```
frame 1860 frame/900= 2.067 start_time= 2 end_time= 2.1 shots= 2
reason=forced_error
```

20. assistant (2026-06-06T06:46:30.652Z)

This is a major finding. Look at rally 1 and rally 2: they BOTH have boxes in frames 180-1260, completely overlapping. Rally 1 has start_time 0.2/end 1.4 (frames 180-1260 ? 0.2/900 to 1260/900=1.4 ? consistent). Rally 2 has start_time **2** but its boxes start at frame 150 (=0.167). So rally 2's start_time attribute (2) does NOT match its frame range (0.167).

Also I see data-entry typos: `shots=1.5` (should be 15), `start\_time=0..2` (typo for 0.2). Let me investigate the overlap structure and the score fields (which all looked empty). And the divisor: 900 = 30fps × 30 (step). Let me verify the score fields and check overlaps systematically.

21. user (2026-06-06T06:46:36.002Z)

```
=== Overlaps between consecutive rally_numbers (n -> n+1) ===
rally 1 [180,1260] vs rally 2 [150,1950]: OVERLAP 1080 frames (1.20s)
total overlapping consecutive pairs: 1
```

```
=== Is rally_number monotonic with frame order? ===
rally 2 fmin=150 < prev fmin 180 NON-MONOTONIC
non-monotonic transitions: 1
```

```
=== Gaps between consecutive rallies (fmax[n] to fmin[n+1]) ===
rally 1->2: gap -1110 frames (-1.23s)
rally 2->3: gap 720 frames (0.80s)
rally 3->4: gap 660 frames (0.73s)
rally 4->5: gap 300 frames (0.33s)
rally 5->6: gap 420 frames (0.47s)
rally 6->7: gap 390 frames (0.43s)
rally 7->8: gap 390 frames (0.43s)
rally 8->9: gap 420 frames (0.47s)
rally 9->10: gap 420 frames (0.47s)
rally 10->11: gap 510 frames (0.57s)
rally 11->12: gap 540 frames (0.60s)
rally 12->13: gap 510 frames (0.57s)
rally 13->14: gap 300 frames (0.33s)
rally 14->15: gap 390 frames (0.43s)
rally 15->16: gap 570 frames (0.63s)
rally 16->17: gap 780 frames (0.87s)
rally 17->18: gap 330 frames (0.37s)
rally 18->19: gap 630 frames (0.70s)
rally 19->20: gap 330 frames (0.37s)
rally 20->21: gap 840 frames (0.93s)
rally 21->22: gap 660 frames (0.73s)
rally 22->23: gap 540 frames (0.60s)
rally 23->24: gap 420 frames (0.47s)
```

22. assistant (2026-06-06T06:46:52.681Z)

Rally 1 and rally 2 overlap by 1080 frames (the entire span of rally 1). This needs deeper investigation ? rally 2 has boxes in 150-1260 AND then 1800-1950. So rally 2 is "split". Let me examine rally 2's full frame distribution and whether rally 1 and rally 2 boxes co-exist on the same frames (two boxes per frame = two players, which is normal) or whether they're genuinely conflated. Let me also check the score fields more carefully and look at the gap between rally 1's claimed time (start_time=0.2) vs rally 2 (start_time=2).

23. user (2026-06-06T06:46:57.224Z)

```
=== Frames with >1 distinct rally_number (conflation) ===
frame 180: rallies [1, 2]
frame 210: rallies [1, 2]
```

```
frame 240: rallies [1, 2]
frame 270: rallies [1, 2]
frame 300: rallies [1, 2]
frame 330: rallies [1, 2]
frame 360: rallies [1, 2]
frame 390: rallies [1, 2]
frame 420: rallies [1, 2]
frame 450: rallies [1, 2]
frame 480: rallies [1, 2]
frame 510: rallies [1, 2]
frame 540: rallies [1, 2]
frame 570: rallies [1, 2]
frame 600: rallies [1, 2]
frame 630: rallies [1, 2]
frame 660: rallies [1, 2]
frame 690: rallies [1, 2]
frame 720: rallies [1, 2]
frame 750: rallies [1, 2]
frame 780: rallies [1, 2]
frame 810: rallies [1, 2]
frame 840: rallies [1, 2]
frame 870: rallies [1, 2]
frame 900: rallies [1, 2]
frame 930: rallies [1, 2]
frame 960: rallies [1, 2]
frame 990: rallies [1, 2]
frame 1020: rallies [1, 2]
frame 1050: rallies [1, 2]
frame 1080: rallies [1, 2]
frame 1110: rallies [1, 2]
frame 1140: rallies [1, 2]
frame 1170: rallies [1, 2]
frame 1200: rallies [1, 2]
frame 1230: rallies [1, 2]
frame 1260: rallies [1, 2]
frames with conflicting rally numbers: 37
```

```
boxes-per-frame histogram: {1: 244, 2: 37}
```

```
rally 2 frames: [150, 180, 210, 240, 270, 300, 330, 360, 390, 420, 450, 480, 510, 540, 570, 600,
630, 660, 690, 720, 750, 780, 810, 840, 870, 900, 930, 960, 990, 1020, 1050, 1080, 1110, 1140,
1170, 1200, 1230, 1260, 1800, 1830, 1860, 1890, 1920, 1950]
```

24. assistant (2026-06-06T06:47:04.263Z)

Now the conflation is crystal clear. For 37 frames (180-1260) there are TWO boxes per frame, one tagged rally 1 and one tagged rally 2. This is NOT two-players-per-frame (the spec attaches rally metadata to a single box). Rally 1 and rally 2 occupy the identical frame span 180-1260, which is impossible for distinct sequential rallies.

Also note rally 2 has a gap: frames 150-1260, then jumps to 1800-1950 (a 540-frame hole from 1290-1770). This strongly suggests an annotation error where rally 1's boxes were duplicated/mislabeled. Let me check whether the two boxes on each conflicted frame have identical coordinates (a true duplicate) or different coordinates.

25. user (2026-06-06T06:47:07.459Z)

```
--- frame 180 ---
  rally=1 xtl=580.78 ytl=451.85 xbr=659.80 ybr=640.63
  rally=2 xtl=1134.10 ytl=434.73 xbr=1286.29 ybr=705.46
--- frame 300 ---
  rally=2 xtl=1134.10 ytl=434.73 xbr=1286.29 ybr=705.46
  rally=1 xtl=580.78 ytl=451.85 xbr=659.80 ybr=640.63
--- frame 1260 ---
  rally=1 xtl=580.78 ytl=451.85 xbr=659.80 ybr=640.63
```

rally=2 xtl=1134.10 ytl=434.73 xbr=1286.29 ybr=705.46

26. assistant (2026-06-06T06:47:16.839Z)

Very revealing. The boxes are at fixed, never-changing positions: rally-1 box always (580.78, 451.85, 659.80, 640.63), rally-2 box always (1134.10, 434.73, 1286.29, 705.46). These boxes do NOT track any player ? they're static markers placed once and never moved across 37+ frames.

This is a "marker box" workflow: the vendor uses a stationary bounding box purely as a carrier for the rally metadata attributes, not for spatial localization. That means:

1. The boxes are useless as spatial GT (they don't track players/shuttle).
2. Rally 1 and rally 2 markers were both left active over frames 180-1260 ? likely rally 1's marker wasn't deleted/closed before rally 2 started, OR (more likely given the interpolation mode) these came from interpolated tracks.

Wait ? mode is "interpolation" but export is per-image. In CVAT interpolation, a track's box persists (interpolated) across all frames between keyframes. Let me verify the static-box hypothesis across ALL rallies, which confirms these are interpolated tracks flattened to images.

27. user (2026-06-06T06:47:20.969Z)

```
rally | nboxes | n_distinct_coords | box_moves?
```

1		37		2		2 positions
2		44		1		STATIC
3		14		1		STATIC
4		11		1		STATIC
5		9		1		STATIC
6		13		1		STATIC
7		20		2		2 positions
8		4		1		STATIC
9		10		1		STATIC
10		7		1		STATIC
11		10		1		STATIC
12		5		1		STATIC
13		10		1		STATIC
14		7		1		STATIC
15		8		1		STATIC
16		26		2		2 positions
17		4		1		STATIC
18		19		1		STATIC
19		6		1		STATIC
20		21		1		STATIC
21		6		1		STATIC
22		9		1		STATIC
23		9		1		STATIC
24		9		1		STATIC

rallies with a single static box position: 21 / 24

28. assistant (2026-06-06T06:47:27.389Z)

Confirmed: 21 of 24 rally "boxes" are completely static (single position across all frames). The 3 that have "2 positions" are exactly the ATTR_VARIES rallies (1, 7, 16). These boxes are metadata carriers, not object localizations. Let me examine rally 7 and 16 to understand the "2 positions" + varying-attr pattern ? likely two separate marker placements got the same rally_number, or a second rally's data leaked in.

29. user (2026-06-06T06:47:30.272Z)

```
=== rally 7 ===
```

```
pos ('570.82', '468.75'): frames 5730-6000 (n=10) st=6.36 et=6.66 shots=2 reason=winner
pos ('444.34', '483.93'): frames 6420-6690 (n=10) st=7.13 et=7.43 shots=4 reason=winner
```

```
=== rally 16 ===
```

```
pos ('919.78', '416.89'): frames 12330-12540 (n=8) st=13.7 et=13.93 shots=2 reason=winner
pos ('945.07', '433.33'): frames 12870-13380 (n=18) st=14.3 et=14.86 shots=8 reason=other
```

30. assistant (2026-06-06T06:47:40.128Z)

Now it's fully clear. Rally 7 and rally 16 each contain TWO physically distinct rally segments that were both mis-tagged with the same rally_number:

- "Rally 7" = two segments: [5730-6000, st=6.36/et=6.66, shots=2, winner] AND [6420-6690, st=7.13/et=7.43, shots=4, winner]. The second segment is a separate rally that should have its own number.
- "Rally 16" = [12330-12540, st=13.7, shots=2, winner] AND [12870-13380, st=14.3, shots=8, other].

So the TRUE rally count is higher than 24. The per-box `start\_time/end\_time` are internally consistent with `frame/900` at the SEGMENT level, confirming the formula. The rally_number field is unreliable. Let me now verify the formula holds at the per-box granularity across ALL boxes (the rigorous test), and confirm the divisor is exactly 900.

31. user (2026-06-06T06:47:44.703Z)

```
segment-level test: st==fmin/900, et==fmax/900
key | fmin | fmin/900 | st | fmax | fmax/900 | et | OK?
rally1 fmin=240 fmin/900=0.267 st=0.2 | fmax=240 fmax/900=0.267 et=1.4 MISMATCH ds=0.067
de=1.133
rally2 fmin=150 fmin/900=0.167 st=2.0 | fmax=1950 fmax/900=2.167 et=2.1 MISMATCH ds=1.833
de=0.067
rally3 fmin=2670 fmin/900=2.967 st=2.9 | fmax=3060 fmax/900=3.400 et=3.4 MISMATCH ds=0.067
de=0.000
rally4 fmin=3720 fmin/900=4.133 st=4.13 | fmax=4020 fmax/900=4.467 et=4.14 MISMATCH ds=0.003
de=0.327
rally6 fmin=4980 fmin/900=5.533 st=5.53 | fmax=5340 fmax/900=5.933 et=5.9 MISMATCH ds=0.003
de=0.033
rally11 fmin=8970 fmin/900=9.967 st=9.96 | fmax=9240 fmax/900=10.267 et=10.23 MISMATCH
ds=0.007 de=0.037
rally12 fmin=9780 fmin/900=10.867 st=10.86 | fmax=9900 fmax/900=11.000 et=10.96 MISMATCH
ds=0.007 de=0.040
rally17 fmin=14160 fmin/900=15.733 st=15.73 | fmax=14250 fmax/900=15.833 et=15.73 MISMATCH
ds=0.003 de=0.103
segments matching frame/900: 19/27
```

32. assistant (2026-06-06T06:47:56.965Z)

My segment splitting was wrong (rally 1 has 2 coords but I split by full 4-tuple incorrectly for the messy ones). Let me reconsider. The key insight: the formula is `time = frame/900`, where $900 = 30 \text{ fps} \times 30 \text{ step}$. The residuals are dominated by data-entry inconsistencies and truncation (start_time entered as truncated 2-decimal). Let me test the formula the right way: check `et ? fmax/900` and `st ? fmin/900` where I round to a consistent precision, and quantify how many segments match within tolerance, treating the known typos separately. Let me also definitively pin the divisor by testing the cleanest cases.

33. user (2026-06-06T06:48:03.008Z)

Testing divisor candidates against $st = \min(\text{frame})/D$ for each st value:

```
D= 900.000: mean|st-minf/D|=0.0756 max=1.8333
D= 899.000: mean|st-minf/D|=0.0882 max=1.8331
D= 901.000: mean|st-minf/D|=0.0828 max=1.8335
D= 900.900: mean|st-minf/D|=0.0816 max=1.8335
D= 899.100: mean|st-minf/D|=0.0869 max=1.8332
D= 900.000: mean|st-minf/D|=0.0756 max=1.8333
```

Direct clean rallies (single static box, attrs constant): st vs fmin/900

34. assistant (2026-06-06T06:48:09.672Z)

D=900 is the best fit. The residual is dominated by the two known bad rallies (rally 2 with st=2 vs frame 150, and rally 1). Let me exclude the contaminated rallies (1, 2, 7, 16) and re-test on the clean ones to nail the formula precision, then do the frame-count reconciliation with

ffprobe.

35. user (2026-06-06T06:48:12.938Z)

Clean rallies (excluding contaminated 1,2,7,16): $st=fmin/900$, $et=fmax/900$
start: $\max|resid|=0.0667$ (these are just 2-dp truncation of $frame/900$)
end: $\max|resid|=0.3267$
i.e. $time = frame/900$ holds to rounding for all 20 clean rallies

36. assistant (2026-06-06T06:48:19.036Z)

Still some larger residuals. The $end=0.3267$ is rally 4. Let me look at exactly which clean rallies still have big residuals ? these may be additional contaminated rallies I haven't flagged, OR cases where the annotator's typed time is more accurate than the sparse box sampling. This matters for "which signal is reliable."

37. user (2026-06-06T06:48:22.110Z)

rally 3: frames [2670, 2700, 2730, 2760, 2790, 2820, 2850, 2880, 2910, 2940, 2970, 3000]
 $fmin=2670$ $fmax=3060$ $st=2.9(fmin/900=2.967)$ $et=3.4(fmax/900=3.400)$ $ncoords=1$
rally 4: frames [3720, 3750, 3780, 3810, 3840, 3870, 3900, 3930, 3960, 3990, 4020] $fmin=3720$
 $fmax=4020$ $st=4.13(fmin/900=4.133)$ $et=4.14(fmax/900=4.467)$ $ncoords=1$
rally 17: frames [14160, 14190, 14220, 14250] $fmin=14160$ $fmax=14250$ $st=15.73(fmin/900=15.733)$
 $et=15.73(fmax/900=15.833)$ $ncoords=1$

38. assistant (2026-06-06T06:48:28.227Z)

These are minor: rally 3 start = 2.9 vs 2.967 (annotator rounded down to 0.1s); rally 4 end = 4.14 vs 4.467 (annotator typed 4.14, likely a typo for 4.47, or hand-judged end before the box track ended); rally 17 end 15.73 vs 15.833 (rounding). These confirm the formula $time = frame/900$ is the generative rule, and the start_time/end_time text fields are lossy hand-transcriptions of it (truncated to ~0.1s, with occasional typos).

The formula is established. Now the critical frame-count reconciliation. Let me run ffprobe on the actual on-disk videos.

39. user (2026-06-06T06:48:28.649Z)

[Request interrupted by user]